

Aliah University

End-Semester Examination (Autumn Semester) - 2024

(For 1st Year 1st Semester MCA)

Paper Name: Database Management Systems Lab

Paper Code: MCAPGPC06

Full Marks: 30

Time: 2 hrs

Date: 09/01/23

1. Consider Employee Table: 15

Emp_no	Emp_Name	Dept	Salary	Branch	Project_id
E100	Robin	Production	60000	Kolkata	P1
E101	Ritesh	Manager	80000	Delhi	P2
E102	Robin	IT	70000	Pune	P1
E103	Susmita	Civil	60000	Kolkata	P2
E104	Amrita	Manager	80000	Mumbai	P4

Answer the SQL Queries:

- Display all the fields of employee table having branch kolkata.
- Retrieve employee number, name and their salary having Project id P1 AND P2.
- Retrieve number of employee working as manager.
- Retrieve distinct number of employees *with distinct name*.
- Calculate the average salary from the table.

2. Create a table Dept. 15

DEPTNO	DNAME	LOC	DEPTHD
D1	EE	KOL	MOHAN
D2	IT	PUN	ZIN
D3	EE	DELHI	TEJOY
D4	ECE	KOL	ANVIOB
D5	ME	DELHI	SAPNA

Perform the following:

- Rename the table Dept as department
- Add a new column PINCODE with not null constraint to the existing table DEPT
- Rename the column DNAME to DEPT_NAME in dept table
- Change the data type of column loc as CHAR with size 10
- Find the DEPTNO whose having O in their DEPTHD name.

19.01.24

Aliah University
End-Semester Examination (Autumn Semester) - 2024
(For 1st Year 1st Semester MCA)

Paper Name: Database Management Systems

Paper Code: MCA PGPC02

Full Marks: 80

Time: 3 hrs

Group-A

Answer all the following questions.

$5 \times 2 = 10$

1.

- a. Define RDBMS.
- b. What is data model? List the types of data model.
- c. State the advantages of DBMS over file processing systems.
- d. Define the phases of two phase locking protocol.
- e. Explain Weak Entity Set with an example.

Group-B

Answer any five questions.

$5 \times 6 = 30$

2. Illustrate different types of joins in SQL. 6
3. Draw an E-R diagram for a hospital using at least five entity sets. 6
4. Explain three schema architectures of DBMS. 6
5. What is functional dependency? Find out the candidate keys and normal form in the following relation R(ABCD) having functional dependencies $AB \twoheadrightarrow C$, $AB \twoheadrightarrow D$, $C \twoheadrightarrow A$, $D \twoheadrightarrow B$. $2+4=6$
6. Define Referential Integrity? Compute the closure of the following set F of functional dependencies for relation schema R(ABCDE): $A \twoheadrightarrow BC$, $CD \twoheadrightarrow E$, $B \twoheadrightarrow D$, $E \twoheadrightarrow A$. $2+2+2=6$
7. Show that the relation $E \twoheadrightarrow AD$ holds in the closure of F.
8. What is Transaction? Discuss the ACID properties of Transaction.

Group-C

Answer any four questions.

$4 \times 10 = 40$

8. Consider the following database schema to write queries in SQL $4+6=10$
Sailor(sid, sname, age, rating)
Boats(bid, bname, bcolor)
Reserves(sid, bid, day)
Insert at least 4 rows of data into each table.

3

 - i) Find the sailors who have reserved a red boat
 - ii) Find the names of the sailors who have reserved at least two boats
 - iii) Find the colors of the boats reserved by 'Mohan'.

9. Discuss serializability and its type. Check the following schedule is Conflict serializable or not. 4+6=

T1	T2	T3
R(X)		
		R(Y)
		R(X)
	R(Y)	
	R(Z)	
		W(Y)
	W(Z)	
R(Z)		
W(X)		
W(Z)		

10. Write short notes on followings

(i) cardinality (ii) BCNF (iii) aggregation (iv) specialization

2.5*4=10

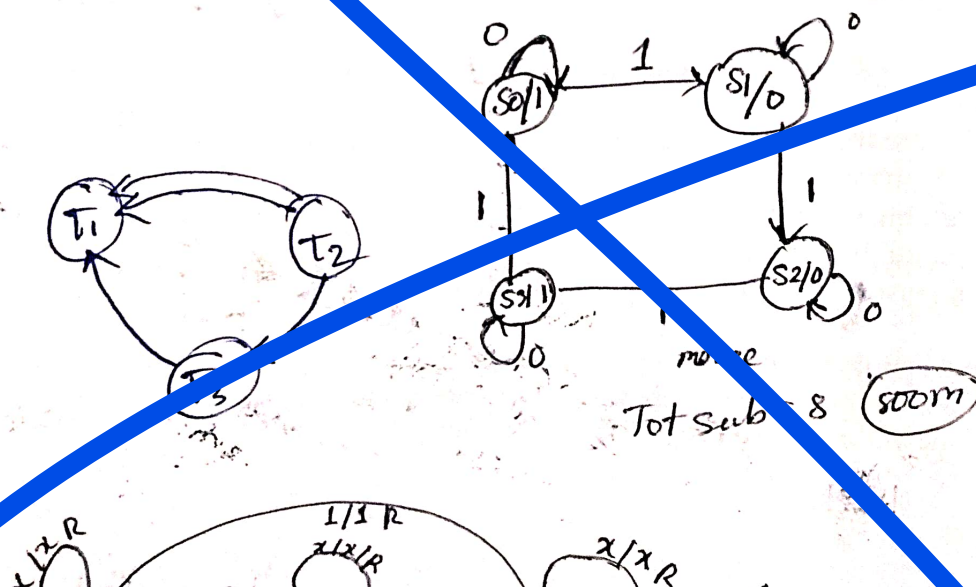
11. Explain lossless and lossy decomposition with an example.

5+5=10

12. What is 2PL Protocol? Build a B tree of order 5 by inserting the following data

D, H, K, Z, B, P, Q, E, A, S, W, C, L, N, Y, M.

10



60
maximum
= 86
Approx = 80

Tot Sub = 8

500m

Aliah University

Odd (Autumn)-Semester Examination - 2023
(For 1st Year 1st Semester BCA Programme)

Paper Name: Computer Organization and Architecture
Paper Code: MCAPGPC03

Full Marks: 80
Time: 1 Hr

Group _____

Answer All questions

(10X1=10)

1. What is the primary function of an ALU (Arithmetic Logic Unit) in a computer?

a) Memory management
b) Performing arithmetic and logical operations

c) Managing input and output devices

2. Which of the following is a primary component of a computer system?

a) RAM

b) ROM

3. What does the term 'clock cycle' refer to in a CPU?

a) The speed at which the CPU executes an instruction

b) The time between two successive clock pulses
c) The time taken to fetch an instruction from memory
d) The time taken to execute an instruction

4. What is the purpose of a 'cache' in a computer system?

a) To store the operating system

b) To act as a backup for the main memory

c) To speed up memory access times

d) To manage peripheral devices

5. In the context of computer organization, what does 'ISA' stand for?

a) Internal System Architecture

b) Integrated Service Architecture

c) Instruction Set Architecture

d) Information Storage Array

6. What is 'pipelining' in the context of CPU design?

a) A method of cooling the CPU

b) A technique for increasing CPU speed

c) A technique for executing multiple instructions simultaneously

d) A method of increasing CPU cache size

7. What is the function of a bus in a computer system?

a) To provide power to components

b) To act as a storage device

c) To transmit data between components

d) To control the system clock

TAOS

8. What does 'RISC' stand for in the context of processor design?

- a) Random Instruction Set Computing
- b) Reduced Instruction Set Computing
- c) Rapid Instruction Sequence Computing
- d) Real-time Instruction Set Computing

9. What is the primary function of the 'control unit' in a CPU?

- a) To perform arithmetic and logical operations
- b) To manage memory operations
- c) To coordinate and control the activities of the other hardware components
- d) To manage input and output devices

10. In the context of computer memory, what does 'RAM' stand for?

- a) Random Access Memory
- b) Readable Archive Memory
- c) Rapid Access Memory
- d) Read-Only Access Memory

Group - B
(Answer any 6 questions)

- 1. Write briefly list the features of each of the generations of computers. 5
- 2. A hierarchical cache main memory subsystem has the following specifications-
i) Cache access time of 60 ns (ii) main memory access time of 300 ns (iii) 80% of memory requests are for read (iv) hit ratio of 0.95 for read access and the write through scheme is used.
Calculate the following: a) Average access time of the memory system considering only memory read cycle b) Average access time of the system both for read and write requests. 2.5+2.5
- 3. Explain the Von Neumann Bottleneck. How is this problem addressed?
- 4. Describe three common cache miss categories. 5
- 5. Explain write through and write back policy. 5
- 6. Point out any three techniques to reduce cache miss. 5
- 7. Draw the memory hierarchy labelling explaining its significance. 5

Group - C
(Answer any 4 questions)

- 1. What is 'Direct Memory Access (DMA)'? How does it improve system performance? (4X10=40)
- 2. Write six major characteristics each of RISC and CISC. 5+5
- 3. Give an example of each of these. 2+2
- 4. Differentiate the different cache mapping techniques. 5
- 5. Describe with example the different types of computers based on performance, size and cost. 5
- 6. Differentiate Sequential, Random, Semi Random and Associative accessing methods with diagrams and examples. 2.5+2.5
- 7. Explain all the modes of data transfer in detail explaining the advantages and disadvantages of each of the modes. 10

DFA

- 1) Input configuration is not allowed.
- 2) Multiple choices are not allowed.
- 3) Corresponding to an input.
- 4) More is not allowed.
- 5) Digital computers are deterministic.
- 6) Designing and understanding is difficult.

NFA

Stochastic feature is not associated with real computers.

21/01/24

Autumn 2023 - 24

End Semester Examination (Autumn Semester) 2023 - 24

(For 1st Year 1st Semester Mathematics)

Paper Name: Computational Mathematics

Paper Code: MATHGP04

Marks: 80

Time: 3 hours

Answer any eight of the following questions.

[8 x 10 = 80]

1. a) Find the inverse of the following matrix by Gauss Elimination and Gauss Jordan methods.

$$\begin{pmatrix} 1 & 2 & 2 \\ 2 & 1 & 2 \\ 2 & 2 & 1 \end{pmatrix}$$

(2 x 5 = 10)

2. a) What is the difference between interpolation and extrapolation? Give suitable examples. What is a formula?

b) From the following table estimate the production of a factory in 1987 using Lagrange interpolation formula.

Year:	1974	1978	1982	1990
Production:	25	60	80	170

3. Find the root of the equation $x^3 - 3x^2 + 5 = 0$ by Regula Falsi method correct up to 3 places after the decimal point.4. Determine the value of $f(x)$ when $x = 42$ by Newton's forward and backward difference interpolation formula.

x:	30	35	40	45
f(x):	10	15	20	25

5. a) Deduce Simpson's 1/3 rule for numerical integration.

b) Evaluate $\int_0^1 (x^2 + x) dx$, taking $h = 0.125$ and 0.0625 , respectively using Romberg's Integration.6. a) Calculate the approximate value of $\int_0^2 (e^{x^2}) dx$ by Composite Trapezoidal Rule by using 10 intervals.b) Calculate the approximate value of $\int_{0.2}^{1.4} (x - x \ln e) dx$ with $h = 0.2$ by using Simpson's 3/8th rule. (5 + 5)

7. Find the Adjoint of the matrix in question 1 and find its inverse. Which of the matrices is (are) orthogonal?

8. What is a singular matrix? Find the trace, determinant and rank of the matrix in Question 1. (1 + 1 + 4)

9. Find all the measures of central tendency, range and interquartile range, variance and standard deviation for the following test scores recorded in a class: 95, 81, 61, 72, 86, 88, 76, 92, 68, 67, 91, 82, 75, 84, 71, 80.

10. a) A card is drawn from a full deck of 52 cards. What is the probability that the card drawn will be either an Ace or a King or a black card?

b) What is the probability that we roll a pair of unbiased 6-sided dice and either get at least one 3 or an odd sum when we add the outcomes of dice together?

c) State the Bayes' Theorem. If we draw a card from a deck of playing cards, and then without replacing it, draw a second card, what is the probability that you will get two Queens in a row? (2 + 3)

Aliah University

Odd-Semester (2022-23) Examination

(For 1st Semester)

Paper Name: Formal Language and Automata Theory

Paper Code: MCAEPC05

Total Marks: 80

Time: 3 Hrs

Answer

Group-A

[5x2=10]

- (i) Define finite automata.
- (ii) Why is context free grammar called "context-free"?
- (iii) When machine is more powerful between NFA and DFA?
- (iv) In which tape of the input tape is writable?
- (v) Write down any two finite automata with output.

Group-B

[5x6=30]

- (i) Give an example of a regular language which is not a finite automata and vice versa. Explain with an example. (3+3)
- (ii) Give a diagrammatic representation of pushdown automata. Explain each of the component. (3+3)
- (iii) Design a finite automata over $\{a, b\}$ which contains at least one "a" and one "b". (6)
- (iv) What are the different acceptance criteria for pushdown automata? Discuss with examples. (6)
- (v) What is a normal grammar? Give examples of regular and context-free grammar. (2+4)
- (vi) Design a pushdown automata which accepts the odd palindromes over $\{a, b\}$. (6)
- (vii) NPDA is more powerful than DPDA. Discuss with example. (6)

Group-B

(Answer any four)

[4x10=40]

3. Answer the following:

(8+2)

- (a) Discuss about Chomsky hierarchy of formal languages.
- (b) What is the configuration of pushdown automata?

4. Draw FA for the following REs (if not possible then justify the reason)

(4x2)

- (a) $(a+b)^*abb$
- (b) Set of all even palindromes over $\{0,1\}$
- (c) $a^*b^*c^* + (a+b)^* + a^*b^*$
- (d) $(0+1+101)^*$ (1000^*)
- (e) Set of all strings over $\{0,1\}$ which has at least one '0'

(4x2.5)

5. Define with suitable example:

- (a) Transition function of NFA
- (b) Different acceptance criteria for pushdown automata
- (c) Right and left linear grammar
- (d) State of Turing machine

6. Answer the following:

(2+2)

- (a) Define pushdown automata
- (b) Design a Turing machine which finds 2's complement of a binary number.

7. Provide correct justification (any two)

(5+5)

- (a) Why are context free languages called "context free"?
- (b) A non-deterministic pushdown automata is more powerful than deterministic pushdown automata.
- (c) Writing in the input tape can make a powerful model of computation.

Aliah University

Odd Semester Examination (Autumn Semester) 2023-24
(For 1st year 1st semester MCA)

Paper Name: Discrete Mathematics and Graph Theory
Paper Code: MCAPGPC01

Full Marks: 80
Time: 3 hours

Group A

Answer all the questions

(5X2=10)

1. Define isomorphism between two graphs.
2. If $A = \{1, 2, 3\}$, $B = \{3, 4, 5\}$ and $C = \{0, 2, 3\}$, find $(A \cap B) \times C$.
3. State the Pigeon-Hole Principle.
4. Define Euler path and Hamiltonian path.
5. How many edges are there in a graph with 10 vertices each having degree 3?

Group B

Answer any 5 questions

(5X6=30)

1. Find the number of integers between 1 to 100 that are not divisible by any of the integers 2, 3, 5 or 7.
2. Prove that the relation R is an equivalence type in the set $P = \{3, 4, 5, 6\}$ given by the relation $R = \{(p, q) : |p - q| \text{ is even}\}$.
3. How many permutations can be made out of the letters of the word "BASIC"? How many of these
(1) Begin with B?
(2) End with C?
(3) B and C occupy the end places.
4. Show that a simple graph G is Eulerian if and only if all its vertices have even degree.
5. Prove that a simple graph with n vertices and k components can have at most $(n - k)(n - k + 1)/2$ edges.
6. How many ways are there to assign five different jobs to four different employees if every employee is assigned at least one job?

Group C

Answer any 4 questions

(4X10=40)

1. a) Using Generating Function method, solve the recurrence relation

$$a_{n+2} - 2a_{n+1} + a_n = 2^n \text{ where, } n \geq 0, a_0 = 2 \text{ and } a_1 = 1$$

- b) Solve the following IDE

$$a_n = 8a_{n-1} + 10^{n-1} \text{ with the initial conditions } a_0 = 1 \text{ and } a_1 = 9 \text{ [5+5]}$$

2. Prove that :

$$1 \cdot 2 \cdot 3 + 2 \cdot 3 \cdot 4 + \dots + n(n+1)(n+2) = \frac{n(n+1)(n+2)(n+3)}{4}$$

b) A committee is formed consisting of one representative from each of the 50 states in the United States, where the representative from a state is either the governor or one of the two senators from that state. How many ways are there to form the committee? [5+5]

3. Explain the following operations on graphs with the help of examples: i) union ii) intersection iii) ring sum iv) fusion v) deletion [5x2]

4. a) A bowl contains 10 red balls and 10 blue balls. A woman selects balls at random without looking at them.

i) How many balls must she select to be sure of having at least three balls of the same color?

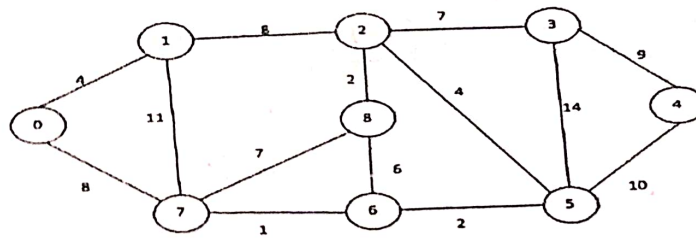
ii) How many balls must she select to be sure of having a least three blue balls?

b) A large software development company employs 100 computer programmers. Of them, 45 are proficient in Java, 30 in C#, 20 in Python, six in C# and Java, five in C# and Python, and just one programmer is proficient in all three languages above. Determine the number of computer programmers that are not proficient in any of these three languages. [5+5]

5.

a) Define a spanning tree of a graph.

b) Write Kruskal's algorithm for finding the minimum spanning tree of a graph. Using the algorithm, find the minimum spanning tree of the given graph.



[2+3+5]

2025
SURE